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A guide to novice for proper selection of the components of drone for specific applications

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ABSTRACT

Since the introduction of drone technology, it has caught the attention of researchers across the world for utilizing it in different areas. But as it is known the same drone cannot be used for all the applications as each and every application has its own unique challenges and requirements. The existing literature emphasizes the use of sophisticated drones for certain applications. But if a novice wants to prepare a tailor made drone for a specific application it is a difficult task for them. This work is an attempt to address the challenges and requirements of drones namely for, agriculture, healthcare, firefighting, and food delivery applications. Which in turn shall be helpful in identifying the right components required for an application? It begins with the statement of various applications of drones and thereby defining the requirements for each application. As per the requirements vigorous calculations are imparted to choose the right combination of electronic components keeping minimal cost in mind. This work could guide any amateur drone enthusiasts to proceed with making their own drone for recreational or commercial purposes. Even the drone entrepreneurs can build their enterprise just by using the output of this manuscript with their expertise or research and development team through proper unmanned vehicle lab facilities.

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1. Introduction

The usage of drones in the modern day has widely increased due to the availability and accessibility of microelectronics that is less in weight and fulfills all our purpose. The name drone is the slang word used for the technical term unmanned aerial vehicles [1]. The range of drones that can be used for commercial purpose could vary from micro drone (i.e. of 30 cm*30 cm*30 cm) to mega drone (i.e. of 300 cm*300 cm*300 cm) [2]. Depending on the mission and application, the size may vary as well as the avionic components necessary for that would be varying. The major hardware and electronic components and hardware used for making drones are tabulated in Table 1 [3].

The drone has found its applications in almost all areas namely: logistics, land surveying, construction, maintenance and demolition, mining, disaster management, surveillance, defense, healthcare, agriculture, firefighting, food delivery, forensics, offshore plants aerial inspections, railways, underwater observation, and marine archeology, etc. The review given below comprises the use of drones in agriculture, healthcare, firefighting, and food delivery:

Agriculture Drone: Various methods and materials to choose and build drones for purposes like crop monitoring and sprinkling system for agriculture applications are described [4]. Also, various hardware components were listed that could be used in such applications. Saha et al. (2018) suggested that drones with IoT can be used for precision agriculture with ease and efficiency [5]. To execute agriculture tasks in a precise manner that may increase yield and produce the best crop quality, Puri et al. (2017) have identified the following drones: Honeycomb AgDrone System, DJI Matrice 100, DJI T600 Inspire 1, Agras MG-1- DJI, EBEE SQ- SenseFly, Lancaster 5 Precision Hawk, and SOLO AGCO Edition [6]. Use of a specific camera (i.e., mobile camera, IR, NIR, and hyperspectral) with a drone was suggested for the crop analysis. [7].

Healthcare Drone: Rosser et al. (2018) present a detailed review of public health and medical surveillance. Here, the emphasis was on using a drone for blood samples transportation from place to place for saving precious life [8]. During Covid-19 supply of

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Table 1
List of components with their purpose.

S. No:	Components	Purpose
1.	Frame	The main platform holds all the electronic components intact.
2.	BLDC Motors	Motors that can produce huge thrust with the least weight.
3.	Propellers	Helps in producing the Lift when rotated with motors.
4.	ESC	Controls the speed of the motors.
5.	LiPO Battery	Has high discharge rate that would help in speed.
6.	Transmitter & Receiver	Sends and Receives commands from pilot to the drone.
7.	Flight Controller	Takes the input commands from Transmitter and controls the various sensors and actuators of drones.

medical aids was a crucial task. The drone was used in the inventory and logistics management of medical aids for improving the efficiency of supply chains [9]. Advanced health care delivery has been revolutionized by medical drones which were inaccessible to the patients earlier [10]. Nyaaba, and Ayamga (2021) have discussed potential benefits, challenges, and healthcare supplies transported by drones [11].

Fire Fighting Drone: Barua et al. (2020) demonstrated a prototype of a fire extinguishing ball thrower quad copter; where some experimental objects have been used to represent the fire-extinguishing balls for future fire extinguishing purposes [12]. For firefighting applications, drones are used for detection, intervention observation, and also for post-fire surveillance [13]. Ausonio et al (2021) have proposed an approach for forest firefighting by employing a swarm of 100 drones that douse the fire by the continuous sprinkling of extinguishing liquid [14]. Alappatt et al. (2021) have proposed a drone that makes use of suitable material other than plastic/fiber for firefighting that doesn't fail at higher temperatures [15].

Food Delivery Drone: Benarbia and Kyamakya (2021) discussed recent drone-based logistics, with a shared autonomous drone package delivery system for parcel delivery [16]. Food delivery drones require a network of charging stations in order to support the business as these drones are power-hungry devices [17]. Hwang and Choe (2019) have analyzed the perceived risk of drone food delivery services and identified the three risks associated with it such as time, performance, and psychological risk which has a negative impact on drone food delivery services [18]. The drones can be used in the production, safety, testing, and warehouse management of food [19].

Currently, varieties of drones are available in the market and the price of the drone varies from few thousands to few lac Rupees or more, depending upon the features/functionalities offered by the drones. For a new entrant/novice in this field it becomes difficult to select the right/ appropriate drones for specific applications amongst the available drones. After rigorous literature review we could not find a literature which specifically suggests how to build an application oriented cost optimized drone(s) with proper selection of components. In the light of the aforementioned problem, we have come up with an idea of preparing a cost effective drone by selection of an appropriate/suitable components for a given application. In this work, we have chosen to develop drone using minimal, cost effective and appropriate components for applications namely, agriculture, healthcare, firefighting, and food delivery.

The rest of the manuscript is structured as follows: Section-2 presents selection of drones based on the application. Section-3 emphasizes the selection of the right components for a drone followed by necessary calculations for the selection of drones based on various payloads in section-4: The conclusion is presented in section-5.

2. Selection of drones based on application

The specific requirements of the drone as per the application are discussed in detail in the following subsections.

Agriculture Drone: Farmer-friendly drones have become sensitized recently after the awareness program government has conducted on various platforms. Such kinds of drones would be having huge payload lifting capabilities with advanced stabilized platforms. The sensors and avionics of such drones would be in the position to withstand all kinds of weather conditions.

Necessary requirements for agriculture Drone: For fertilizer, a 10 to 15-liter spraying tank is required with high spraying efficiency. The body of the drone needs to be sturdy and agile in any kind of motion. Eight spray nozzles would be required for equal distribution and extensive coverage. The flow rate of the nozzle designed should be around 4 to 8 L per minute. The spray needs to ensure strong penetration and exceptional drift prevention. These 8 nozzles could be placed below the 8 thrusters themselves as the downwash from the thrusters further increase the flow velocity of spray. Independent variable frequency control and spraying are done by 8 sets of solenoid valves. The agriculture drone does need a horizontally opposed six-cylinder double plunger pump for the strong spraying power. The spray width would be about 8 m.Keeping these design requirements, the agriculture drone should be capable of spraying 10 Acres in an hour.[20].

Some of the extra features that could be added to the drone to make it a next-generation drone would be adding a radar system, obstacle avoidance system, and camera system. For the choice of the radar system, a spherical radar system would be preferred as it perceives obstacles, as well as everything around it in any environmental conditions, also comes with advanced viewing angles be it dust or light interference. To ensure the safety of drones from any collision or obstacle hit the obstacle avoidance system should be automatic along with adaptive flight functions. For improved awareness dual front pilot view (FPV) cameras are apt. These equipped FPV cameras would provide clear views both in forward as well as in rearward. The advantage of keeping in both forward and rearward will be that the drone doesn't require performing yaw motions to get a complete view of the surroundings. The camera should be accompanied by searchlights to improve the drone's night vision capabilities to perform better in nighttime operations. For such kind of operation, the requirement for the battery would be around 25000mAh which should give at least 1000 flight cycles. The charging should be instant charging with no external cooling mechanism. The battery should be water-resistant, corrosion preventive layers, and circuit board potting protection. A battery station would be necessary for charging the batteries. It should possess at least 6800 W of charging power and should charge the battery within10 minutes.[21].

Healthcare Drone: After the effect of the pandemic, the usage of healthcare purpose drones has become popular. The healthcare drone would carry a payload that could be medicines, blood, vaccine kits, test samples, small lab equipment, or accessories. Even vital organs such as the heart, kidneys, lungs, and eyes could be transported through drones for surgeries; provided they are maintained in a payload case with essential storage conditions. The application of a healthcare drone can be extended for spraying disinfectant across the cities by using the same principle that was imparted in the agriculture drone session.

Necessary requirements for healthcare Drone: For the healthcare drone, speed would be the major requirement.

So rather providing the conventional multi-copter, the design should be hybrid vertical take-off and landing (VTOL) drone. In such types of drones, the take-off and landing would be similar to that of multi-copters as it requires small take-off or landing space and cruising would be similar to that of fixed-wing drones as it achieves the highest possible forward cruising speeds. The cruising speed should be 100 kmph. Payload capabilities play a vital role in design when it comes to fixed-wing, as the effects would lead to major changes in design. Still for a payload of 2 kg medical supplies, within the flight range of 160 km, the wingspan needed would be 3 m.

The weight of the fuselage for such small payloads would range from 6 to 8 kg and the weight of wings could be 2–4 kg. It is also necessary to attach a parachute with a payload if the payload needs to be dropped in some remote areas or no land area. A Global Positioning System (GPS) module needs to be equipped with the drone for tracing the position of a drone as well as to guide the drone to the destination.

Fire Fighting Drone: Fire and safety departments could make use of drones to extinguish the fire, in that way casualties of humans could be nullified during the firefighting incidence. Usage of thermal cameras embedded with drones advances this application. It can scan hotspots, which would be able to show us the visibility even through smokes as well as in low light conditions. Flashlights in drones would guide the firefighters to approach the scene in dark and smoky environments.

Necessary requirements for firefighting Drone: The Drone should possess 8 BLDC motors with a suitable combination of Electronics Speed controllers. To extinguish the fire, a fireball of diameter 15.2 cm could be used. The weight of the ball is almost 1.5 kg. When it gets in contact with flame, it gets activated within 3 to 10 s. The camera DJI Zenmuse XT would be the best choice for such kinds of applications, thermal imaging at 640/30 FPS or a 336/30 FPS with high-sensitivity 50 mK provision. This sensitivity provides very accurate temperature measurements which would be ideal for analytics and telemetry. The DJI gimbals could be used for better stability and control; they provide very smooth, clear pictures and 360 degrees of seamless rotational movement. The appropriate battery could be a six-cell Li-Po battery of 1200 mAh 15c current capacity and 22.2 V. Double Lume Cube 1.0 waterproof LED lights would be needed and can be operated at least for 2 h.[22].

2. Food Delivery Drone: The idea of a food delivery drone would be the most successful in the market. It would attract more customers and would replace the delivery partners of the most online food business. Such drones also could be used effectively during pandemic scenarios. It can even provide food to the people at disaster-prone times and especially during floods.

Necessary requirements for food delivery Drone: To withstand thrust and vibrations, a frame with great strength would be required. It is preferred to use foldable arms with carbon fiber frames. In heavy wind times, GPS flying mode would give the drone stable flight. Six-axis octa copter frame would be suitable for great efficiency, better weight lifting capacity, and stability. Forty Five minutes of flying time with a 6 kg payload could be achieved with a proper selection of batteries. With the fast charging and discharging capability of LiPO battery, the drone could achieve ranges up to 5 km radius with 200 m vertical altitudes. In case of low battery or loss of signals from the remote, the drone needs to return to the controller station, so one button return to home option needs to be installed. Variable opening flaps could be used to release delivery hooks. For longer life and low maintenance, a powerful and efficient propulsion system is a must. It should take only 30 s to catch the satellite and be ready to go. Attractive LEDs would help in night flights preferable with an anti-collision system.

3. Selection of right components for a drone

Batteries: LiPo or Lithium Polymer has a high watt/weight rating and deliver up lot of current so would be the proper choice for a given application. Parameters to be looked for selection of a battery are as follows:

- Number of cells: The choice of a number of cells, like 1, 2, or 3 cells (or 11.1 V) would determine the voltage of the battery. The nominal voltage is 3.7 V per cell.
- Capacity: Storage capacity is indeed the major parameter. It would be noted in milli ampere-hours or mAh as denoted in the battery. A battery with 2000 mAh refers to a battery with 2 Ah. The endurance of the battery depends on it. If 1 Ampere is drawn from the battery, it would last for 2 h, if 2 Ampere is drawn, the battery would last for 1 h, and if 4 Ampere is drawn, the battery would last for ½ hour or 30 min.
- C Rating: It is defined as the measurement of the current with which a battery is charged and discharged at.
- It measures the power of the battery that can be delivered without dropping the maximum of its voltage.
- The amount of current in Ampere that could be delivered would be dependent on the battery capacity.
- The current delivery is just the product of C rating and battery capacity. Example: current delivered by 20C rating battery with 1000mAh would be 20 A.
- Discharging the Battery: The LiPo batteries can be destroyed if they are being discharged completely. The threshold limit for the battery to discharge is 3.0 V, which should not happen. If discharge goes beyond this value, the battery would catch fire. As per the 80% rule, batteries should never been discharged more than 80% of their capacity rating. For example, a 1000mAh battery should never be discharged more than 800mAh.
- Charging the Battery: A charger that displays a mAh rating that would come in handy. Measure the flight time of the drones. If the drone flew for 5 min and then the battery is charged such that 500 mAh is charged back to the battery, let the drone draws 100mAh per minute, then it can fly up to 8 min at 1C charge.

Propeller: The propellers have two identification numbers at their top. The identification number indicated first is the diameter of the propeller in inches and the second identification number is the pitch of the propeller in inches. These identification numbers are usually engraved at top of the propeller. (Say Example: 10*4.5).

- Diameter of the propeller– This identification number indicates the diameter of the propeller measured from tip to tip in inches. The thrust generated is depicted by the diameter of the propeller.
- Pitch of the propeller –This identification number indicates the distance that a propeller could travel forward in one complete revolution. A pitch with higher numbers would be suitable for higher speeds and generally would lack efficiency at less speed. A pitch with lower numbers would have a better grip and a faster acceleration but with a lower top speed. (Note: Identification numbers would be facing forward always. It is essential to check the airfoil shape of the propeller; the edge with a rounded surface needs to be facing forward. If that edge faces backward, it would produce the least thrust with more current being drawn. This mistake is common among beginners.)

Motor: The possible propulsion system to drive the propellers could be fuel engines, brushed motors, and brushless motors. But the Electric brushless motor has been used widely due to its ease of availability and environment friendly.

The brushless DC electric motors are of two types:

- Inrunner type - Only the shafts of the motor would spin. Generally used in faster drones.
- Outrunner type - Has more torque and most commonly used, the whole body outside of the motor would spin. These motors come in all sizes and shapes.
- The major parameters that need to be understood while choosing the motor:
- Size of the motor: Selection of the size of the motor could be confusing for beginners as it depends on the manufacturers. In general, the motor is identified by the stator size. It would be printed on the motor like: "2210–12". In this identity number, the first two digits would give the diameter in mm, while the next two digits would give the height in mm, and finally after the dash, the last number would give the number of "turns" for each winding in the motor. (See Fig. 1).
- Power Rating: Care needs to be given while identifying the power rating as wattage would be rated differently for various voltages. Example: A power of 150 W at a 15 V voltage would give quality that of 100 W at 10 V as the current which would play a crucial role. The logic behind the size of the motor goes like this – the larger motors could produce more power without overheating than smaller motors.
- kV Rating: The kV should not be confused with kilo-volt but it's a small k that represents constant. In general, the kV stands for the number of revolutions per minute / per volt. For example, a 1000 kV motor would spin about 10,000 RPM on a voltage of 10 V. The kV rating is usually calculated while the motor has been not loaded. The kV rating does not deal with the power of the motor. It just represents the speed of the motor.

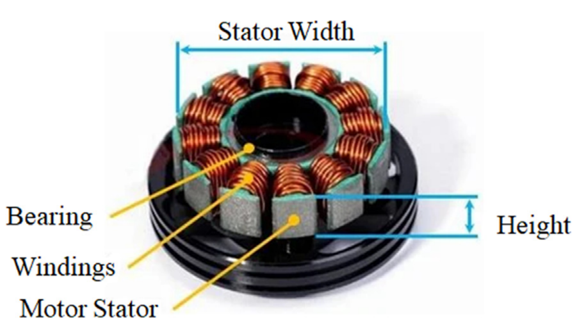

Fig. 1. Brushless DC Motor Terminologies [23].

ESC (Electronic Speed Controller): The major consideration for choosing an ESC would be the current, the voltage, and the necessity of a BEC.

- Current Rating of ESC: It gives the amps that ESC can deliver continuously. Most of the ESCs would list a 'max burst' (usually 10 s or less.)
- The rating doesn't mean the amount of power it would pump out to the electric motor; rather it is the amount of current that the ESC can govern before it reaches the threshold.
- The ESC with more amps capacity doesn't mean anything but just to weigh more.
- Voltage Rating of ESC: The ESC does have a voltage range that is usually operated within and it would be printed on the cover of ESC. The same threshold voltage should not be exceeded or the ESC would not respond or would be destroyed.
- Necessity of BEC: The Battery Elimination Circuit would power the Receiver used to control the various parts in the drone. Unless the receiver is powered from a separate battery, the need for ESC with built-in BEC arises.
- Choosing the right speed controller: Check out the amp (or watt) rating of the motor and thereby using the 80% rule, the speed controller can be chosen. An ESC of 12 amps or more would be required for the motor rated for 10 amps.
- Choosing the right Motor size and kv: In the motor specifications, it mentioned how much thrust motor can produce according to that we can select the motor. Motor size is the stator diameter and stator length.
- Choosing the right ESC: After selecting the motor, we can go for the ESC selection in the motor specifications it is mentioned that how much amps motor draws from the battery according to that ESC should be selected.

4. Necessary calculations for the selection of drones based on various payloads

Single Motor Thrust Calculation: Formula for Single Motor Thrust = (Total required thrust × 2) / drone configuration. A thrust measuring rig was made to measure out the maximum thrust a motor can produce. There is a wooden plank that was attached to a simple weighing machine. The motor is mounted on the wooden plank with the propellers fastened be reversed, when the motors were driven, the thrust generated is acting towards the weighing machine. The scale to indicate thrust generated by the motor and propeller would be displayed in the weighing machine. Various motors, propellers, and battery combinations were tested on the

Table 2
Thrust obtained from different motors with propellers and ESC with Battery combinations.

Motor size (L*B)(mm)	Motor (kv)	Propeller size (inch)	ESC Current (A)	ESC Voltage (V)	Single motor thrust(gm)
19.1*28.9	2550	6*4	32.9	16.2	1406
26.4*41.8	580	14*4.8	16	15	1500
48*35.3	790	13*6.5	29.4	12	1755
36*35.3	1450	11*7	51.5	11.8	1884
48*35.5	1150	9*6	54.6	11.4	2241
26*34	925	9*5	50.5	25	2333.333333
26*34.4	925	9*5	50.5	24.5	2992
34*49	900	13*6.5	57.8	13.3	3271
45*42.8	800	12*6	58.6	20.7	3635
26.5*86.8	135	30*10.5	21	24	4333.333333
45*60.7	490	18*6.1	62	25	5600
45*60.7	490	18*6.1	62	25	5750
33.5*69	340	22*6.6	57.8	24	6000
26.1*87.1	190	29*9.5	44	24	7250
46.3*60	220	19*10	60	44	8750
36*57.6	490	16*5.4	83.2	33	9259

Table 3
Power required calculated for the considered range of payload through the chosen components.

Sr. No.	Payload Weight (gm)	Total require thrust (gm)	Drone configuration	Single motor thrust (gm)	Motor size (L*B) (mm)	Motor (kV)	Propeller size (inch)	ESC Current (A)	ESC Voltage (V)	Power Required (W)
1	2000	3000	4	1500	26.4*41.8	580	14*4.8	16	15	46517.6
2	5000	7000	6	2333.3333	26*34	925	9*5	50.5	25	114211.80
3	10,000	13,000	6	4333.3333	26.5*86.8	135	30*10.5	21	24	162407.79
4	15,000	18,000	6	6000	33.5*69	340	22*6.6	57.8	24	136794.24
5	20,000	23,000	8	5750	45*60.7	490	18*6.1	62	25	161893.10
6	25,000	29,000	8	7250	26.1*87.1	190	29*9.5	44	24	182956.3
7	30,000	35,000	8	8750	46.3*60	220	19*10	60	44	358563.33

Table 4
Optimum Cost for basic electronic components with estimated total cost for various payload weights.

Payload (gm)	2000	5000	10,000	15,000	20,000	25,000	30,000
Drone Configuration	4	6	6	6	8	8	8
Motor price (Rs)	17,600	28,800	35,000	42,000	56,000	56,000	72,000
ESC price (Rs)	4000	12,000	17,000	24,000	36,000	32,000	36,000
Propeller price (Rs)	1800	3000	4200	5400	9600	9600	12,000
Battery price (Rs)	5000	8000	28,000	40,000	50,000	50,000	75,000
Flight Controller price (Rs)	20,000	33,800	33,800	33,800	56,400	56,400	26,400
Frame price (Rs)	5900	10,700	10,700	10,700	49,250	49,250	49,250
Total	54,300	96,300	128,700	155,900	257,250	253,250	270,650

rig to obtain the values of thrust. The Table 2 represents the values obtained from the test.

From the Table 2, the optimum motor, propeller, and battery that could be used for various applications were chosen by testing various combinations.

The Table 3 gives us the Thrust and Power required for the chosen range of payloads along with the correct specification of the chosen optimum motor, propeller, and battery combination.

The cost mentioned in Table 4 is just for the basic parts used in a drone to perform the flight. The actual cost would go further high depending on the various sensors, GPS Modules, and Transmitter-Receiver Modules. As well the accessories required to support the flight controllers will change depending on the mission and company of the manufacturer.

A detailed explanation of various applications of drones in the commercial sector along with the technical requirements that would make any amateur drone enthusiasts take drone venture at ease. In the selection of the right components for drones, the detailed parameters and instructions to choose the components with those parameters have been discussed in section-3. In the calculation part, the calculation of a single motor thrust for every payload rise was obtained. Then using bench tests various motors and propellers combinations were tested to obtain different values of thrust. Through the thrust required formulas, we can obtain the power required, and hence appropriate battery would be chosen. By choosing the correct motor, propeller, electronic speed controller, and battery, the optimization of the cost was performed and tabulated in Tables 2-4, which would be the main outcome of this manuscript. The authors envisage that this manuscript would be a starting document for a novice who wishes to learn and explore drone technology.

5. Conclusion

The selection of a drone is a most difficult task as it includes monetary terms and has to perform the desired task which cannot be compromised. Thus significant specification for a drone for applications like healthcare, fire-Fighting, Agriculture, and food Delivery Drones have been presented in this manuscript. With the list of electronic components specified in Table 1, any amateur drone enthusiasts can purchase the mentioned sensors and equipments from the market and start building their own personal drone. The application oriented technical requirements mentioned the section-2 would allow anyone to build the commercial drone for various missions. To make the things little bit easier, this manuscript have explained the science behind each and every component used in the drones along with their technical details in the section-3 that would assist anyone to choose the right components. Stating the method of choosing the right component and thereby optimizing the total cost for the entire mission is a scoring process for any drone researcher. The range of payload starts from 2 kg to 30 kg and depends on specific applications and depending on the payload capacity the cost is going to vary. Further, by considering the extra accessories and various varieties of products the range of cost can vary from 1 lac Indian rupees to 20 lac Indian rupees.Table 2.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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